

Robust Fall Recovery for Armless Bipedal-Wheeled Robots via Force-Guided Learning

Haidong Hou, Zhangguo Yu, Tao Han, Hengbo Qi, Khaleel Ghazal, Yu Zhang, Yidong Du, Xuechao Chen, and Fei Meng

Abstract—Fall recovery is critical for autonomous legged locomotion. Existing methods have demonstrated that some legged robots, such as humanoids and quadrupeds, are capable of fall recovery from diverse postures by utilizing arms or coordinating multi-legs to generate support forces. Without arms or other legs to provide supportive assistance, a bipedal-wheeled robot must rely solely on the actuation of its legs, making recovery particularly difficult. To address this, we introduce FTSR (Force-guided Teacher-student framework with Stage-wise Rewards). The force-guided method constructs an external auxiliary force during simulation training that correlates directly with the robot's real-time height, explicitly formulating this force as an optimizable constraint. Through constrained reinforcement learning, the policy is guided toward reducing force dependency gradually and increasing the body height, developing internal recovery strategies despite having no arms for support. Height-progressive stage-wise rewards progressively structure posture stabilization during recovery and transition to sustained locomotion, integrated with teacher-student architecture distilling privileged knowledge of force effects and recovery dynamics. After simulation training, the policy is deployed on a physical armless bipedal-wheeled robot and extensively evaluated. Experiments confirm robust and reliable fall recovery under diverse challenging conditions, demonstrating strong environmental adaptability and motion robustness, while maintaining full post-recovery motion capability. The framework also generalizes effectively to a high-DOF humanoid, confirming its practical generalizability. The project page is available at <https://2350575870.github.io/force-guided.github.io/>.

Index Terms—Bipedal-Wheeled Robots, Force-Guided Learning, Constrained Policy Optimization, Stage-Wise Reward.

I. INTRODUCTION

BIPEDAL-WHEELED robots combine legged locomotion's terrain adaptability with wheeled mobility's efficiency, drawing significant research interest [1], [2]. While some significant progress has been achieved in legged navigation on complex terrain [3], [4], [5], [6], [7], legged robots remain vulnerable to falls under real world uncertainties, often resulting in arbitrary, unpredictable postures.

The fall recovery for legged robots has been historically studied through model-based strategies. Early approaches, applicable across various robotic forms, relied on executing

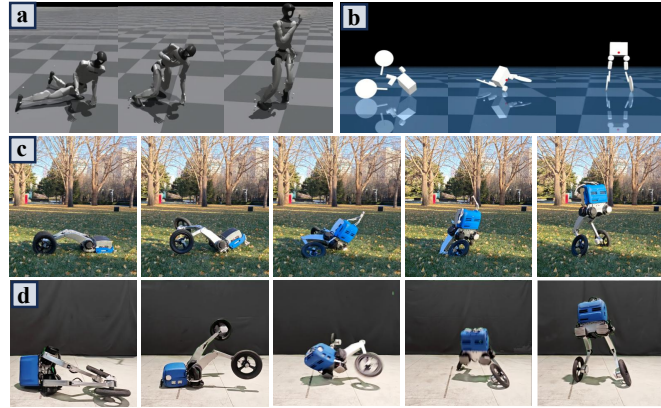


Fig. 1. **Simulation and physical deployment results.** Subfigures (a) and (b) show simulated recovery of a high-DOF Unitree robot and a bipedal-wheeled robot, respectively, while (c) and (d) present successful real-world recovery of the latter under varying environments and initial poses.

fixed, precomputed trajectories often derived from motion capture or trajectory optimization [8]. While offering a deterministic solution, such methods are highly sensitive to the initial pose and lack the robustness and generalization needed for the diverse falling configurations encountered in practice.

To address these limitations, research has pivoted towards data-driven frameworks using reinforcement learning (RL) and imitation learning. Some approaches [9] initiated training from simplified initial poses but replaced model-based optimization with end-to-end policy learning, substantially improving both the success rate and motion robustness of the recovery process. Relevant research [10], [11] advanced to address the more challenging scenarios of recovery from multiple poses and across varied terrains. To date, some studies [12], [13], [14] often employ staged training frameworks or incorporate curriculum learning, which decomposes the complex recovery task into progressively difficult phases to facilitate stable policy acquisition. Notably, HoST [15] defines multiple categories of rewards with distinct weighting coefficients to form a composite learning objective while using a force curriculum that applies and gradually phases out upward external assistance. This combined approach enhances the training process by refining policy quality and accelerating convergence. In parallel, imitation learning offers a complementary pathway for generating complex motor skills, enabling robots to acquire more human-like and natural recovery motions by leveraging expert demonstrations or video data [16], [17], [18].

In contrast, armless bipedal-wheeled robots face significant challenges in recovery, as they must rely solely on their

基于力引导学习的无臂双轮足机器人鲁棒跌倒恢复

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摘要—跌倒恢复对于自主腿式运动至关重要。现有方法已证明, 部分腿式机器人(如人形机器人和四足机器人)能够通过利用手臂或协调多腿产生支撑力, 从多样姿态中实现跌倒恢复。对于没有手臂或其他腿部提供支撑性辅助的双足轮式机器人, 其必须完全依赖自身腿部的驱动, 这使得恢复尤为困难。为解决这一问题, 我们提出了FTSR(力引导师生框架与分阶段奖励)。该力引导方法在仿真训练中构建一个与机器人实时高度直接相关的外部辅助力, 并明确将该力表述为可优化约束。通过约束强化学习, 策略被引导逐步降低力依赖并提升身体高度, 从而在无手臂支撑的情况下发展内部恢复策略。高度渐进式分阶段奖励在恢复过程中逐步构建姿态稳定, 并过渡到持续运动, 同时结合师生架构蒸馏力效应和恢复动力学的特权知识。仿真训练后, 该策略被部署到实体无臂双足轮式机器人上并进行广泛评估。实验证实, 在多种挑战性条件下, 该方法能够实现稳健可靠的跌倒恢复, 展现出强大的环境适应性和运动鲁棒性, 同时保持完整的恢复后运动能力。该框架还能有效泛化至高自由度人形机器人, 验证了其实际泛化性。项目页面见 <https://2350575870.github.io/force-guided.github.io/>。

索引词——双足轮式机器人, 力引导学习, 约束策略优化, 分阶段奖励。

BIPEDAL-WHEELED 机器人结合了腿式运动的地形适应性与轮式移动的高效性, 引起了广泛的研究兴趣 [1], [2]。然而

尽管在复杂地形上的腿式导航已取得一些重大进展, [3], [4], [5], [6], [7], 腿式机器人在现实世界的不确定性下仍然容易摔倒, 常常导致任意不可预测姿态。

腿式机器人的跌倒恢复历来通过基于模型的策略进行研究。早期方法适用于各种机器人形态, 依赖于执行

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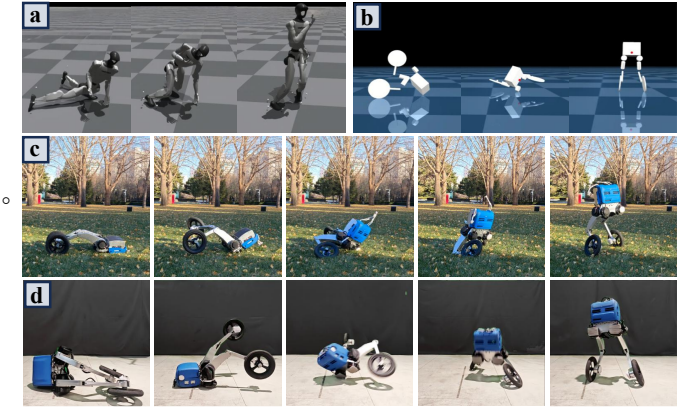


图 1. 仿真与实物部署结果。子图 (a) 和 (b) 分别展示了高自由度人形机器人和双足轮式机器人的仿真恢复过程, 而 (c) 和 (d) 则展示了后者在不同环境和初始姿态下成功完成真实世界恢复的结果。

固定的、预计算轨迹, 这些轨迹通常源自动作捕捉或轨迹优化 [8]。虽然提供了确定性的解决方案, 但此类方法对初始姿态高度敏感, 且缺乏应对实际中遇到的各种跌倒构型所需的鲁棒性和泛化能力。

为解决这些局限性, 研究已转向采用强化学习 (RL) 和模仿学习的数据驱动框架。一些方法 [9] 从简化的初始姿态开始训练, 但将基于模型的优化替换为端到端策略学习, 显著提高了恢复过程的成功率和运动鲁棒性。相关研究 [10], [11] 进一步推进, 以应对从多种姿态和跨多变地形恢复等更具挑战性的场景。迄今为止, 一些研究 [12], [13], [14] 通常采用分阶段训练框架或引入课程学习, 将复杂的恢复任务分解为逐步困难的阶段, 以促进稳定的策略获取。值得注意的是, HoST [15] 定义了多类具有不同权重系数的奖励, 形成复合学习目标, 同时使用力课程来施加并逐步取消向上的外部辅助。这种组合方法通过优化策略质量和加速收敛来增强训练过程。与此同时, 模仿学习为生成复杂运动技能提供了一条补充路径, 使机器人能够利用专家示范或视频数据 [16], [17], [18] 获得更类人的、自然的恢复运动。

相比之下, 无臂双足轮式机器人在恢复方面面临重大挑战, 因为它们必须仅依靠

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legs for actuation without the support of arms or additional legs. A dynamic recovery from a specific posture has been demonstrated on bipedal-wheeled robots in non-peer-reviewed video content [19]. However, existing methods, which are fundamentally reward-based, often converge prematurely to local optima when applied to armless bipedal-wheeled robots. This issue is particularly acute in complex terrains where special circumstances prevent the robots from shifting to a specific initial posture, reducing recovery success rates or causing complete failure. These methods, when training constraints become excessive and convergence conditions overly demanding, often converge prematurely to local optima, termed dead points, resulting in policy training failure. This problem is exacerbated on bipedal-wheeled robots where limited joint actuation is concentrated in the lower body without upper-body support, making premature convergence severely undermine both learning efficiency and real-world robustness.

To address these issues, we propose the FTSR framework. Its core force-guided learning constructs an external auxiliary force during simulation that correlates directly with the robot's real-time height, explicitly formulated as optimizable hard constraints [20] rather than a decaying curriculum. This constraint-driven approach prevents the policy from converging directly to useless local optima, enabling robust and adaptive fall recovery for armless bipedal-wheeled robots. To achieve stable posture refinement and enable subsequent locomotion after fall recovery, we introduce height-progressive stage-wise rewards that progressively update the target base height based on height thresholds. This key design element ensures stable training, while the stage-wise structure [21] guides the robot through posture refinement and subsequent locomotion after standing up. Supplemented by teacher-student architecture [22] adapted to capture privileged force parameters, FTSR achieves efficient autonomous recovery with real-world validation, and shows transferability potential to high-DOF humanoid robots, as shown in Fig. 1.

The main contributions of this work are as follows:

- The force-guided method we propose constructing height-correlated external auxiliary forces as optimizable constraints is proposed to restrict actions to physically feasible recovery trajectories and enable robust fall recovery for armless bipedal-wheeled robots without supportive assistance, readily extending to high-DOF humanoids.
- Height-progressive stage-wise rewards enable stable training while guiding the robot through posture refinement and subsequent locomotion after recovery.
- For the first time, fall recovery for an armless bipedal-wheeled robot from multiple randomized postures and diverse terrains has been achieved in real-world scenarios, validating the robustness and adaptability of the approach.

II. METHOD

To address premature convergence to dead points in armless bipedal-wheeled robot fall recovery, we propose the Force-guided Teacher-student framework with Stage-wise Rewards (Sec. II-B, Alg. 1). Unlike curriculum-based methods that heuristically phase out external assistance, we introduce force-guided learning that formulates auxiliary forces and torques

as optimizable constraints within CPO (Sec. II-A), guiding the policy toward reducing force dependency gradually as it develops recovery strategies without arm support (Sec. II-C). Meanwhile, height-progressive stage-wise rewards automatically transition from posture refinement to locomotion based on batch height statistics, ensuring stable training (Sec. II-D).

A. Constrained Markov Decision Process

A Constrained Markov Decision Process (CMDP) [23] is an extension of the Markov Decision Process, which must satisfy additional constraints (such as resource limitations or safety requirements) while optimizing the long-term cumulative reward. In this paper, we employ a CMDP to achieve progressive policy training from strong guidance to autonomy through tightenable external-force constraints. A CMDP can be formalized as a tuple $\langle S, A, P, R, \gamma, C, d \rangle$, where S denotes the state space, A is action space, P is state transition probability, $R(s_t, a_t, s_{t+1})$ denotes the immediate reward function, γ is discount factor, $C = C_i(s_t, a_t, s_{t+1})_{i=1}^m$ represents m constraint costs incurred when executing an action (each constraint function $C_i : S \times A \times S \rightarrow \mathbb{R}$ is defined similarly to the reward function), and $d = \{d_1, \dots, d_i, \dots, d_m\}$ denotes the allowable upper limits for each constraint cost. Our objective is to learn an optimal policy $\pi^* : S \rightarrow A$ that maximizes the expected discounted return while satisfying all constraints over the trajectory τ :

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t, s_{t+1}) \right] \quad (1)$$

s.t. $J_{C_i}(\pi) \leq d_i \quad \forall i \in \{1, \dots, m\}$

In the above formulation, $J_{C_i}(\pi)$ represent the expected return of the i -th constraint under policy π , respectively, defined as follows:

$$J_{C_i}(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t C_i(s_t, a_t, s_{t+1}) \right] \quad (2)$$

To solve the aforementioned optimization problem, [20] and [24] proposed the Constrained Policy Optimization (CPO) algorithm, which integrates constraints into the trust-region framework to enable larger update steps:

$$\begin{aligned} \pi_{k+1} &= \arg \max_{\pi \in \Pi_\theta} \mathbb{E}_{s \sim d^{\pi_k}} [A^{\pi_k}(s, a)] \\ \text{s.t. } J_{C_i}(\pi_k) + \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d^{\pi_k}} [A_{C_i}^{\pi_k}(s, a)] &\leq d_i \quad \forall i \quad (3) \\ \bar{D}_{KL}(\pi \| \pi_k) &\leq \delta \end{aligned}$$

where Π_θ is the set of parameterized policies with parameters θ , d^π denotes the state visitation distribution under policy π , defined by $d^\pi(s) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t P(s_t = s | \pi)$, $\bar{D}_{KL}(\pi \| \pi_k) = \mathbb{E}_{s \sim d^{\pi_k}} [D_{KL}(\pi \| \pi_k)[s]]$ represents the expected KL-divergence between policies, $\delta > 0$ is the maximum step size, $A^{\pi_k}(s, a)$ is the advantage function for the reward, and $A_{C_i}^{\pi_k}(s, a)$ is the advantage function for the i -th constraint.

腿部驱动, 而没有手臂或额外腿部的支撑。在非同行评审的视频内容中, 已展示了双足轮式机器人从特定姿态进行动态恢复 [19]。然而, 现有的方法从根本上基于奖励机制, 在应用于无臂双足轮式机器人时, 往往过早收敛到局部最优。这一问题在复杂地形中尤为突出, 因为特殊情况会阻止机器人转换到特定的初始姿态, 从而降低恢复成功率或导致完全失败。当训练约束过于严苛且收敛条件要求过高时, 这些方法常常过早收敛到局部最优, 即所谓的死点, 导致策略训练失败。这一问题在双足轮式机器人上更为严重, 因为有限的关节驱动集中在下半身, 缺乏上半身支撑, 使得过早收敛严重削弱了学习效率和现实世界鲁棒性。

为了解决这些问题, 我们提出了FTSR框架。其核心的力引导学习在仿真过程中构建了一个与机器人实时高度直接相关的外部辅助力, 并明确地将其表述为可优化的硬约束 [20], 而非衰减课程。这种约束驱动方法防止策略直接收敛到无用的局部最优, 从而实现了无臂双足轮式机器人稳健且自适应的跌倒恢复。为了实现稳定的姿态调整并在跌倒恢复后能够进行后续运动, 我们引入了高度渐进式分阶段奖励, 该奖励根据高度阈值逐步更新目标基准高度。这一关键设计元素确保了训练的稳定性, 而分阶段结构 [21] 引导机器人完成姿态调整以及站立后的后续运动。辅以适用于捕获特权力参数的师生架构 [22], FTSR实现了高效的自主恢复, 并通过现实世界验证, 同时展现出向高自由度人形机器人迁移的潜力, 如图1所示。

本研究的主要贡献如下:

- 我们提出的力引导方法通过构建与高度相关的外部辅助力作为可优化约束, 将动作限制在物理可行的恢复轨迹内, 使无臂双足轮式机器人无需支撑性辅助即可实现鲁棒跌倒恢复, 并可轻松扩展到高自由度人形机器人。
- 高度渐进式分阶段奖励能够在引导机器人完成姿态调整及恢复后的运动的同时, 确保训练的稳定性。
- 我们首次在现实世界场景中实现了无臂双足轮式机器人从多种随机姿态和多样化地形中的跌倒恢复, 验证了该方法的鲁棒性和适应性。

II. 方法

为了解决无臂双足轮式机器人在跌倒恢复过程中过早收敛到死点的问题, 我们提出了带分阶段奖励的力引导师生框架 (第II-B节, 算法1)。与基于课程的方法启发式地逐步取消外部辅助不同, 我们引入了力引导学习, 将辅助力和力矩

作为约束策略优化中的可优化约束 (第II-A节), 引导策略在无手臂支撑的情况下逐步降低力依赖, 同时发展恢复策略 (第II-C节)。同时, 高度渐进式分阶段奖励基于批次高度统计自动从姿态调整过渡到运动, 确保训练稳定 (第II-D节)。

A. 约束马尔可夫决策过程

约束马尔可夫决策过程 (CMDP) [23]是马尔可夫决策过程的扩展, 在优化长期累积奖励的同时, 必须满足额外的约束 (如资源限制或安全要求)。在本文中, 我们采用CMDP, 通过可收紧的外部力约束, 实现从强引导到自主的渐进式策略训练。CMDP可以形式化地表示为一个元组 $\langle S, A, P, R, \gamma, C, d \rangle$, 其中 S 表示状态空间, A 是动作空间, P 是状态转移概率, $R(s_t, a_t, s_{t+1})$ 表示即时奖励函数, γ 是折扣因子, $C = C_i(s_t, a_t, s_{t+1})_{i=1}^m$ 表示执行动作时产生的 m 约束代价 (每个约束函数 $C_i : S \times A \times S \rightarrow \mathbb{R}$ 的定义与奖励函数类似), $d = \{d_1, \dots, d_i, \dots, d_m\}$ 表示每个约束代价的允许上限。我们的目标是学习一个最优策略 $\pi^* : S \rightarrow A$, 在满足轨迹 τ 上所有约束的同时, 最大化期望折扣回报:

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t, s_{t+1}) \right] \quad (1)$$

s.t. $J_{C_i}(\pi) \leq d_i \quad \forall i \in \{1, \dots, m\}$

在上述公式中, $J_{C_i}(\pi)$ 分别表示策略 π 下第 i 个约束的期望回报, 定义如下:

$$J_{C_i}(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t C_i(s_t, a_t, s_{t+1}) \right] \quad (2)$$

为解决上述优化问题, [20]和 [24]提出了约束策略优化 (CPO) 算法, 该算法将约束整合到信任域框架中, 以实现更大的更新台阶:

$$\begin{aligned} \pi_{k+1} &= \arg \max_{\pi \in \Pi_\theta} \mathbb{E}_{s \sim d^{\pi_k}} [A^{\pi_k}(s, a)] \\ \text{s.t. } J_{C_i}(\pi_k) + \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d^{\pi_k}} [A_{C_i}^{\pi_k}(s, a)] &\leq d_i \quad \forall i \quad (3) \\ \bar{D}_{KL}(\pi \| \pi_k) &\leq \delta \end{aligned}$$

其中 Π_θ 是参数为 θ 的参数化策略集合, d^π 表示策略 π 下的状态访问分布, 定义为 $d^\pi(s) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t P(s_t = s | \pi)$, $\bar{D}_{KL}(\pi \| \pi_k) = \mathbb{E}_{s \sim d^{\pi_k}} [D_{KL}(\pi \| \pi_k)[s]]$ 表示策略之间的期望KL散度, $\delta > 0$ 是最大步长, $A^{\pi_k}(s, a)$ 是奖励的优势函数, $A_{C_i}^{\pi_k}(s, a)$ 是第 i 个约束的优势函数。

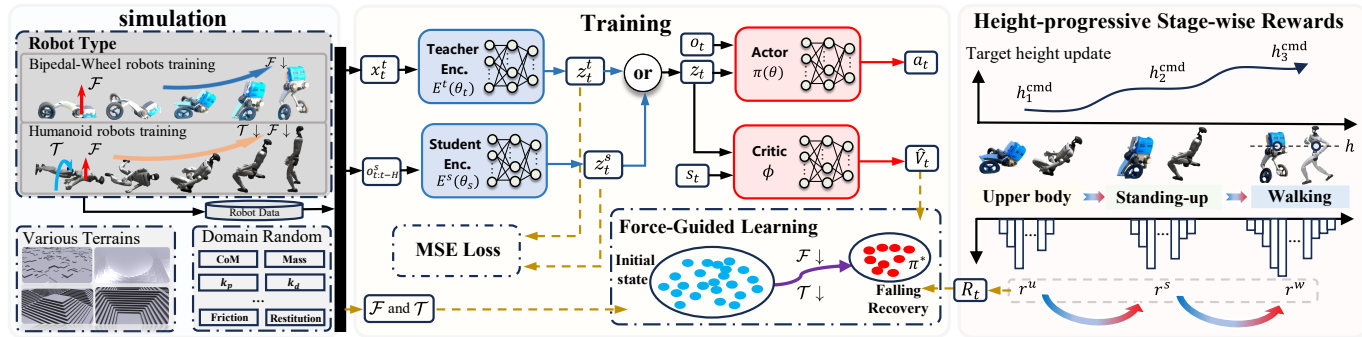


Fig. 2. **Force-guided fall-recovery training framework.** The framework supports either bipedal-wheeled or humanoid robots, selected from 'Robot Type'. Training data flows from the selected robot through "Robot Data" to generate teacher encoder observation x_t^t and student history observation $o_{t:t-H}^s$, which feed E^t and E^s respectively to produce latents z_t^t and z_t^s . These latents are then selected by an "or" module, with z_t^t for teacher-group agents and z_t^s for student-group agents, to form z_t concatenated with o_t and s_t for shared Actor-Critic networks (π and ϕ) outputting a_t and $\hat{V}_t$. During training, yellow arrows indicate training signals: z_t^t and z_t^s supervise E^s via MSE loss, while $\mathcal{F}$, $\mathcal{T}$, R_t , and $\hat{V}_t$ update E^t , π , and ϕ to evolve policy toward π^* by progressively reducing $\mathcal{F}$ and $\mathcal{T}$ via Force-Guided Learning. As robot base height increases, R_t transitions from r^u to r^w when agents exceeding height thresholds reach two-thirds of the agents, synchronizing target height and rewards with actual height progression.

B. Force-Guided Reinforcement Learning Framework

The overall framework is illustrated in Fig. 2. Armless bipedal-wheeled robots lack upper-body support during fall recovery, making them prone to convergence to dead points under pure reward optimization. This motivates formulating external auxiliary forces $\mathcal{F}$ and torques $\mathcal{T}$ as optimizable constraints rather than decaying curricula, enabling the policy to gradually reduce dependency on external assistance while discovering physically feasible recovery trajectories. Such constraint-guided exploration necessitates stable posture refinement across phases, which we achieve through height-progressive stage-wise rewards that auto-transition based on batch height statistics rather than fixed durations.

Realizing these mechanisms requires balancing privileged information with proprioceptive-only execution, leading to a teacher-student architecture where teacher encoder E^t processes x_t^t comprising contact forces, base height, and full bodies state, while student encoder E^s learns equivalent representations from history $o_{t:t-H}^s$. This history concatenates observations o_t over past H steps, where each $o_t \in \mathbb{R}^{34}$ includes angular velocity, gravity projection, user commands, joint positions, velocities, and previous actions. An "or" module routes latent z_t^t or z_t^s to shared Actor-Critic networks by agent group, producing z_t concatenated with current o_t and $s_t \in \mathbb{R}^{188}$ containing height map and robot base height. The networks output actions a_t and values $\hat{V}_t$, with parameters in Table I and component designs in Sec. II-C and Sec. II-D.

C. Force-Guided Fall Recovery Training

Force curriculum heuristically phases out assistance, causing collapse before stable standing is acquired and convergence to dead points under excessive constraints. We instead propose force-guided learning, formulating forces and torques as optimizable constraints within CPO, reshaping their functional forms to enable gradual reduction tied to policy improvement. The expressions for force $\mathcal{F}$ and torque $\mathcal{T}$ are given in Equation (4).

TABLE I
NETWORK ARCHITECTURE

Models	Input	Hidden layer	Output
Actor $\pi(\theta)$	z_t, o_t	[512, 256, 128]	a_t
Critic ϕ	z_t, s_t	[512, 256, 128]	$\hat{V}_t$
Privileged Enc. $E^t(\theta_t)$	x_t^t	[256, 128]	z_t^t
Student Enc. $E^s(\theta_s)$	$o_{t:t-H}^s$	[256, 128]	z_t^s

$$\begin{aligned} \begin{bmatrix} \mathcal{F} \\ \mathcal{T} \end{bmatrix} &= \left(1 - e^{-\mu(h_i^{\text{cmd}} - h)}\right) \text{sat}_{[0,1]}(t_{\text{coeff}}) \\ &\quad \times \begin{bmatrix} F_{\max} n \\ T_{\max} \log(R_{\text{tag}} R^{-1}) \end{bmatrix} \end{aligned} \quad (4)$$

In the formulation, μ is the height coefficient, which determines the extent to which height influences the force and torque. h_i^{cmd} denotes the target height of the center of mass (CoM), while h represents the current CoM height. The time coefficient is given by $t_{\text{coeff}} = 1 - t/t_{\text{tag}}^{\text{step}}$, where t is the current time step and $t_{\text{tag}}^{\text{step}}$ is the step at which external assistance ends, and $\text{sat}_{[0,1]}(\cdot)$ denotes a saturation function that clips the parameter to the range $[0, 1]$. $F_{\max}$ indicates the maximum external force applied to the robot, and $T_{\max}$ is the maximum applied torque. R_{tag} is the target rotation matrix generated from the target Euler angles, and R is the rotation matrix derived from the current Euler angles. The force axis is defined by the vector $n = [0 \ 0 \ 1]^T$. The notation $\log(\cdot)$ represents the logarithmic map on the $so(3)$ group, which maps the relative rotation matrix $R_{\text{tag}} R^{-1}$ to an instantaneous rotation vector $\mathbf{w} \in \mathbb{R}^3$: $\mathbf{w} = \log(R_{\text{tag}} R^{-1})$. The direction of this vector corresponds to the axis of rotation, and its 2-norm equals the rotation angle in radians.

However, the goal is not merely to apply forces, but to train a policy that adapts to them, learning recovery with progressively reduced intervention. Inspired by CPO, we formulate the force and torque as two constraint conditions: $C_1 = \mathcal{F}$ with $J_{C_1}(\pi) \leq d_1$ and $C_2 = \mathcal{T}$ with $J_{C_2}(\pi) \leq d_2$, and integrate them into the training process. We then incorporate these two

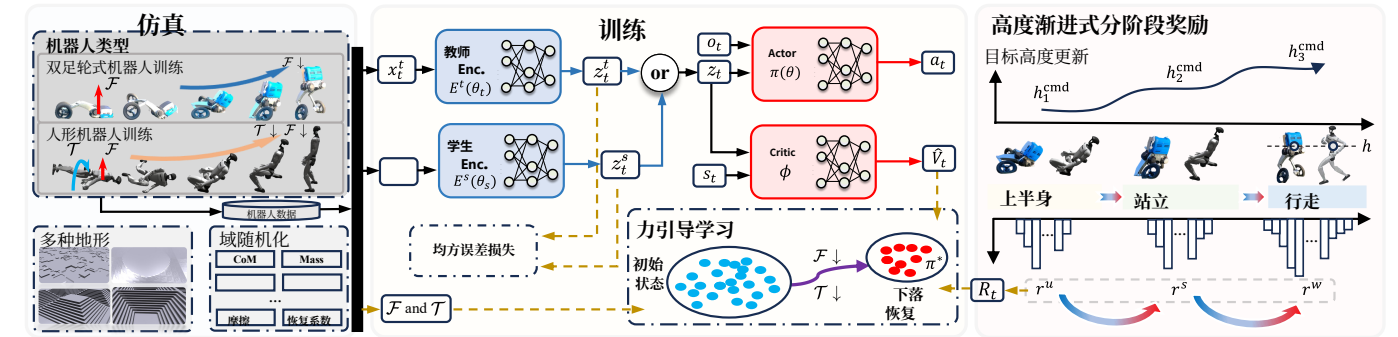


图2. 力引导跌倒恢复训练框架。该框架支持从“机器人类型”中选择双足轮式机器人或人形机器人。训练数据从所选机器人通过“机器人数据”流向教师编码器观测 x_t^t 和学生历史观测 $o_{t:t-H}^s$ ，分别输入 E^t 和 E^s 以生成潜变量 z_t^t 和 z_t^s 。这些潜变量随后由“或”模块选择，其中 z_t^t 用于教师组智能体， z_t^s 用于学生组智能体，形成 z_t 并与 o_t 和 s_t 拼接，供共享的Actor-Critic网络（ π 和 ϕ ）输出 a_t 和 $\hat{V}_t$ 。训练期间，黄色箭头表示训练信号： z_t^t 和 z_t^s 通过均方误差损失监督 E^s ，而 $\mathcal{F}$ 、 $\mathcal{T}$ 、 R_t 和 $\hat{V}_t$ 更新 E^t 、 π 和 ϕ ，通过力引导学习逐步减少 $\mathcal{F}$ 和 $\mathcal{T}$ ，使策略向 π^* 演化。随着机器人基座高度增加，当超过高度阈值的智能体达到智能体总数的三分之二时， R_t 从 r^u 过渡到 r^w ，使目标高度和奖励与实际高度进展同步。

B. 力引导强化学习框架

总体框架如图2所示。无臂双足轮式机器人在跌倒恢复过程中缺乏上半身支撑，在纯奖励优化下容易收敛到死点。这促使我们将外部辅助力 $\mathcal{F}$ 和力矩 $\mathcal{T}$ 建模为可优化约束而非衰减课程，使策略能够在发现物理可行的恢复轨迹的同时逐步减少对外部辅助的依赖。这种约束引导的探索需要跨阶段的稳定姿态调整，我们通过基于批次高度统计而非固定时长自动切换的高度渐进式分阶段奖励来实现这一目标。

实现这些机制需要在特权信息与仅本体感觉执行之间取得平衡，从而形成教师-学生架构，其中教师编码器 E^t 处理 x_t^t 包含接触力、基座高度和全身状态的信息，而学生编码器 E^s 从历史数据中学习等效表示 $o_{t:t-H}^s$ 。该历史数据将过去 o_t 步骤中的观测 H 拼接起来，其中每个 $o_t \in \mathbb{R}^{34}$ 包括角速度、重力投影、用户指令、关节位置、速度和先前动作。一个“或”模块根据智能体组将潜变量 z_t^t 或 z_t^s 路由到共享的Actor-Critic网络，生成 z_t 与当前 o_t 和 $s_t \in \mathbb{R}^{188}$ （包含高度图和机器人基座高度）拼接。该网络输出动作 a_t 和值 V_t ，参数见表I，组件设计见第II-C节和第II-D节。

C. 力引导跌倒恢复训练

力课程启发式地逐步取消外部辅助，导致在稳定站立习得之前发生崩溃，并在过度约束下收敛到死点。我们转而提出力引导学习，将力和力矩表述为约束策略优化中的可优化约束，重塑其函数形式，使其能够随策略改进而逐步减小。力 $\mathcal{F}$ 和力矩 $\mathcal{T}$ 的表达式如公式 (4) 所示。

表 I 网络架构

模型	输入	隐藏层	输出
Actor $\pi(\theta)$	z_t, o_t	[512, 256, 128]	a_t
Critic ϕ	z_t, s_t	[512, 256, 128]	$\hat{V}_t$
特权编码器 $E^t(\theta_t)$	x_t^t	[256, 128]	z_t^t
学生编码器 $E^s(\theta_s)$	$o_{t:t-H}^s$	[256, 128]	z_t^s

$$\begin{aligned} \begin{bmatrix} \mathcal{F} \\ \mathcal{T} \end{bmatrix} &= \left(1 - e^{-\mu(h_i^{\text{cmd}} - h)}\right) \text{sat}_{[0,1]}(t_{\text{coeff}}) \\ &\quad \times \begin{bmatrix} F_{\max} n \\ T_{\max} \log(R_{\text{tag}} R^{-1}) \end{bmatrix} \end{aligned} \quad (4)$$

在该公式中， μ 是高度系数，它决定了高度对力和力矩的影响程度。 h_i^{cmd} 表示质心 (CoM) 的目标高度，而 h 表示当前质心高度。时间系数由 $t_{\text{coeff}} = 1 - t/t_{\text{tag}}^{\text{step}}$ 标签给出，其中 t 是当前时间步， $t_{\text{tag}}^{\text{step}}$ 标签是外部辅助结束的时间步， $\text{sat}_{[0,1]}(\cdot)$ 表示将参数裁剪到范围 $[0, 1]$ 内的饱和函数。 $F_{\max}$ 表示施加到机器人上的最大外力， $T_{\max}$ 是最大施加力矩。 R_{tag} 是由目标欧拉角生成的目标旋转矩阵， R 是由当前欧拉角推导出的旋转矩阵。力轴由向量 $n = [0 \ 0 \ 1]^T$ 定义。符号 $\log(\cdot)$ 表示在 $so(3)$ 群上的对数映射，它将相对旋转矩阵 $R_{\text{tag}} R^{-1}$ 映射为瞬时旋转向量 $\mathbf{w} \in \mathbb{R}^3$: $\mathbf{w} = \log(R_{\text{tag}} R^{-1})$ 。该向量的方向对应旋转轴，其2-范数等于以弧度表示的旋转角。

然而，目标不仅仅是施加力，而是训练一个能够适应这些力的策略，在逐步减少干预的情况下学习恢复。受约束策略优化的启发，我们将力和力矩表述为两个约束条件： $C_1 = \mathcal{F}$ 与 $J_{C_1}(\pi) \leq d_1$ 以及 $C_2 = \mathcal{T}$ 与 $J_{C_2}(\pi) \leq d_2$ ，并将它们整合到训练过程中。随后，我们将这两个

constraints into Equation (3). Unlike the original CPO formulation that solves constrained quadratic programs, we employ the penalty function method to transform the optimization into a simpler form containing only a KL-divergence:

$$\begin{aligned} & \arg \max_{\pi \in \Pi_\theta} \mathbb{E}_{\substack{s \sim d^{\pi_k} \\ a \sim \pi}} [A^{\pi_k}(s, a)] \\ & + \sum_{i=1}^m \beta_i \left(d_i - \left(J_{C_i}(\pi_k) + \frac{1}{1-\gamma} \mathbb{E}_{\substack{s \sim d^{\pi_k} \\ a \sim \pi}} [A_{C_i}^{\pi_k}(s, a)] \right) \right) \\ & \text{s.t. } \bar{D}_{KL}(\pi \| \pi_k) \leq \delta \end{aligned} \quad (5)$$

In the formulation, $A^{\pi_k}(s, a)$ and $A_{C_i}^{\pi_k}(s, a)$ are computed using Generalized Advantage Estimation (GAE). Prior to gradient calculation using the advantage functions, an additional standardization step is applied to enhance training stability. Equation (5) is optimized via Proximal Policy Optimization (PPO).

To enable the policy model to directly access privileged information, thereby accelerating convergence and improving training stability, we employ a teacher-student model [25]. Its objective function is given by Equation (6):

$$L^{\text{cpo},j}(\vartheta) = \frac{1}{|\mathcal{D}^j T|} \sum_{\tau \in \mathcal{D}^j} \sum_{t=0}^T \min \left(r_t^j \bar{A}_t^j, \text{sat}_{[1-\epsilon, 1+\epsilon]} \left(r_t^j \right) \bar{A}_t^j \right) \quad (6)$$

In this formulation, the superscript $(\cdot)^j$ denotes whether the teacher or the student model is being used ($j = t$ indicates teacher-group parameters, and $j = s$ indicates student-group parameters). The subscript $(\cdot)_t$ denotes the value of the bracketed parameter at time step t . ϑ represents the mixed network weights: for the teacher group, $\vartheta = \theta + \theta^t$; for the student group, $\vartheta = \theta$. T is the trajectory length of the agent-environment interaction, and $\mathcal{D}^j$ is the trajectory data generated by the corresponding policy. r_t^j is the importance sampling ratio, defined explicitly in Equation (7). ϵ is the clipping range parameter in PPO. $\bar{A}_t^j$ denotes the mixed advantage function that combines the value function and constraint functions, as defined in Equation (8).

$$r_t^j = \begin{cases} \frac{\pi(a_t^t | o_t, E^t(x_t^t))}{\pi_{\text{old}}(a_t^t | o_t, E^t(x_t^t))} & j = t \\ \frac{\pi(a_t^s | o_t, E^s(o_{t:t-H}^s))}{\pi_{\text{old}}(a_t^s | o_t, E^s(o_{t:t-H}^s))} & j = s \end{cases} \quad (7)$$

$$\begin{aligned} \bar{A}_t^j &= A_t^j(s, a) - \sum_{i=1}^m \beta_i \\ & \left(J_{t, C_i}^j(\pi_k) + \frac{1}{1-\gamma} \mathbb{E}_{\substack{s \sim d^{\pi_k} \\ a \sim \pi}} [A_{t, C_i}^j(s, a)] \right) \end{aligned} \quad (8)$$

In the formulation, $J_{t, C_i}^j(\pi_k)$ denotes the constraint function value at time step t in the trajectory, and $A_{t, C_i}^j(s, a)$ is the corresponding constraint advantage function. Unlike Equation (5), our training objective aims to drive the external forces and torques acting on the agent to zero; hence, d_i should be

as small as possible. To ensure the mixed advantage function simplifies to the target form $\bar{A}_t^j = A_t^j(s, a)$ as given in Equation (8), we set $d_i = 0$ when $t = t_{\text{tag}}^{\text{step}}$.

The Critic network is trained using a mixed dataset $\mathcal{D} = \mathcal{D}^t \cup \mathcal{D}^s$ and updated by minimizing the mean squared error between its predicted value $\hat{V}_t$ and the target value R_t estimated from trajectory returns:

$$L^{\text{value}}(\phi) = \frac{1}{|\mathcal{D}T|} \sum_{\tau \in \mathcal{D}} \sum_{t=0}^T \left(\hat{V}_t - R_t \right)^2 \quad (9)$$

For the student encoder, we employ the Mean Squared Error (MSE) as the training objective. Unlike the approach in [25], we do not restrict training to trajectories generated solely by the student group; instead, we directly utilize all trajectories obtained from agent-environment interactions for training.

$$L^s(\theta_s) = \frac{1}{|\mathcal{D}T|} \sum_{\tau \in \mathcal{D}} \sum_{t=0}^T \|E^s(o_{t:t-H}^s) - E^t(x_t^t)\|_2^2 \quad (10)$$

Algorithm 1 FTSR Algorithm

- 1: Initialize environment and networks
- 2: Initialize Penalty Factor β_i , Learning Rate $\alpha_{cpo}, \alpha_{mse}$ for training.
- 3: Empty replay buffer $\mathcal{D}^t$ and $\mathcal{D}^s$
- 4: Use ϑ to denote θ and θ_t for brevity
- 5: **for** $0 \leq \text{its} \leq \text{iterations}$ **do**
- 6: Compute $\mathcal{F}$ and $\mathcal{T}$ using Eq. (4)
- 7: $C_i \leftarrow (\mathcal{F}, \mathcal{T})$
- 8: Compute A_t^j using GAE, compute $\bar{A}_t^j$ using Eq. (8)
- 9: Update reward via threshold vector:
- 10: $R_t \leftarrow [r^u, r^w, r^s] \cdot [\mathbb{I}(p=0), \mathbb{I}(p=1), \mathbb{I}(p=2)]^\top$
- 11: where $p = 2 \cdot \mathbb{I}(|S_1| > \frac{2}{3}N) + \mathbb{I}(|S_2| > \frac{2}{3}N)$
- 12: Actor-Critic and Teacher Enc. update:
- 13: $\vartheta \leftarrow \vartheta + \alpha_{cpo} \nabla_{\vartheta} (L^{\text{cpo},t}(\vartheta) + L^{\text{cpo},s}(\theta))$
- 14: $\phi \leftarrow \phi + \alpha_{cpo} \nabla_{\phi} L^{\text{value}}(\phi)$
- 15: **end for**
- 16: **for** epoch $i = 0, 1, \dots$ **do**
- 17: Student Enc. update using MSE:
- 18: $\theta_s \leftarrow \theta_s + \alpha_{mse} \nabla_{\theta_s} (L^s(\theta_s))$
- 19: **end for**

D. Height-progressive Stage-wise Rewards

To address training instability caused by fixed target heights in prior stage-wise reward designs, we adopt height-progressive stage-wise rewards that progressively update target base heights based on agents-level height thresholds. The final reward functions and their weights are listed in Table II. Each layer employs a distinct reward set to guide the robot sequentially toward recovery: **(1) Upper Body Erection** r^u . This initial phase aims to drive the robot's upper body from a prone position to an upright posture, primarily guided by a gravity projection penalty, with the target base height set to h_1^{cmd} . We define agent sets $S_1 = \{i \mid h > h_1^{\text{cmd}}\}$ and $S_2 = \{i \mid h > h_2^{\text{cmd}}\}$ based on height thresholds. **(2)**

约束纳入方程(3)。与原始约束策略优化公式中求解约束二次规划不同，我们采用惩罚函数方法将优化转化为仅包含KL散度的更简单形式：

$$\begin{aligned} & \arg \max_{\pi \in \Pi_\theta} \mathbb{E}_{\substack{s \sim d^{\pi_k} \\ a \sim \pi}} [A^{\pi_k}(s, a)] \\ & + \sum_{i=1}^m \beta_i \left(d_i - \left(J_{C_i}(\pi_k) + \frac{1}{1-\gamma} \mathbb{E}_{\substack{s \sim d^{\pi_k} \\ a \sim \pi}} [A_{C_i}^{\pi_k}(s, a)] \right) \right) \\ & \text{s.t. } \bar{D}_{KL}(\pi \| \pi_k) \leq \delta \end{aligned} \quad (5)$$

在该公式中， $A^{\pi_k}(s, a)$ 和 $A_{C_i}^{\pi_k}(s, a)$ 使用广义优势估计（GAE）计算。在使用优势函数进行梯度计算之前，会应用额外的标准化步骤以增强训练稳定性。公式（5）通过近端策略优化（PPO）进行优化。

为了使策略模型能够直接访问特权信息，从而加速收敛并提高训练稳定性，我们采用了教师-学生模型 [25]。其目标函数由公式（6）给出：

$$L^{\text{cpo},j}(\vartheta) = \frac{1}{|\mathcal{D}^j T|} \sum_{\tau \in \mathcal{D}^j} \sum_{t=0}^T \min \left(r_t^j \bar{A}_t^j, \text{sat}_{[1-\epsilon, 1+\epsilon]} \left(r_t^j \right) \bar{A}_t^j \right) \quad (6)$$

在此公式中，上标 $(\cdot)^j$ 表示使用的是教师模型还是学生模型（ $j = t$ 表示教师组参数，而 $j = s$ 表示学生组参数）。下标 $(\cdot)_t$ 表示括号内参数在时间步 t 处的值。 ϑ 表示混合网络权重：对于教师组， $\vartheta = \theta + \theta^t$ ；对于学生组， $\vartheta = \theta$ 。 T 是智能体-环境交互的轨迹长度，而 $\mathcal{D}^j$ 是由相应策略生成的轨迹数据。 r_t^j 是重要性采样比率，在公式（7）中明确定义。 ϵ 是PPO中的裁剪范围参数。 $\bar{A}_t^j$ 表示结合了价值函数和约束函数的混合优势函数，如公式（8）中所定义。

$$r_t^j = \begin{cases} \frac{\pi(a_t^t | o_t, E^t(x_t^t))}{\pi_{\text{old}}(a_t^t | o_t, E^t(x_t^t))} & j = t \\ \frac{\pi(a_t^s | o_t, E^s(o_{t:t-H}^s))}{\pi_{\text{old}}(a_t^s | o_t, E^s(o_{t:t-H}^s))} & j = s \end{cases} \quad (7)$$

$$\begin{aligned} \bar{A}_t^j &= A_t^j(s, a) - \sum_{i=1}^m \beta_i \\ & \left(J_{t, C_i}^j(\pi_k) + \frac{1}{1-\gamma} \mathbb{E}_{\substack{s \sim d^{\pi_k} \\ a \sim \pi}} [A_{t, C_i}^j(s, a)] \right) \end{aligned} \quad (8)$$

在公式中， $J_{t, C_i}^j(\pi_k)$ 表示轨迹中时间步 t 处的约束函数值，而 $A_{t, C_i}^j(s, a)$ 是相应的约束优势函数。与公式（5）不同，我们的训练目标旨在将作用于智能体的外力和力矩驱动至零；因此， d_i 应当

尽可能小。为确保混合优势函数简化为公式（8）中给出的目标形式 $\bar{A}_t^j = A_t^j(s, a)$ ，我们设置 $d_i = 0$ 当 $t = t_{\text{tag}}^{\text{step}}$ 标签时。

评论家网络使用混合数据集 $\mathcal{D} = \mathcal{D}^t \cup \mathcal{D}^s$ 进行训练，并通过最小化其预测值 V_t 与从轨迹回报估计的目标值 R_t 之间的均方误差来更新：

$$L^{\text{value}}(\phi) = \frac{1}{|\mathcal{D}T|} \sum_{\tau \in \mathcal{D}} \sum_{t=0}^T \left(\hat{V}_t - R_t \right)^2 \quad (9)$$

对于学生编码器，我们采用均方误差（MSE）作为训练目标。与 [25]中的方法不同，我们 not 将训练限制在仅由学生组生成的轨迹上，而是直接利用从智能体-环境交互中获得的所有轨迹进行训练。

$$L^s(\theta_s) = \frac{1}{|\mathcal{D}T|} \sum_{\tau \in \mathcal{D}} \sum_{t=0}^T \|E^s(o_{t:t-H}^s) - E^t(x_t^t)\|_2^2 \quad (10)$$

算法1 FTSR算法

- 1: 初始化环境和网络
- 2: 初始化罚因子 β_i 、学习率 $\alpha_{cpo}, \alpha_{mse}$ 用于训练。
- 3: 清空回放缓冲区 $\mathcal{D}^t$ 和 $\mathcal{D}^s$
- 4: 使用 ϑ 表示 θ ，并使用 θ_t 以简化表达
- 5: **对于** $0 \leq \text{其} \leq \text{次迭代}$ **执行**
- 6: 计算 $\mathcal{F}$ 和 $\mathcal{T}$ 使用公式（4）
- 7: $C_i \leftarrow (\mathcal{F}, \mathcal{T})$
- 8: 计算 A_t^j 使用GAE，计算 $\bar{A}_t^j$ 使用公式（8）
- 9: 通过阈值向量更新奖励：
- 10: $R_t \leftarrow [r^u, r^w, r^s] \cdot [\mathbb{I}(p=0), \mathbb{I}(p=1), \mathbb{I}(p=2)]^\top$
- 11: 其中 $p = 2 \cdot \mathbb{I}(|S_1| > \frac{2}{3}N) + \mathbb{I}(|S_2| > \frac{2}{3}N)$
- 12: 演员-评论家与教师编码器更新：
- 13: $\vartheta \leftarrow \vartheta + \alpha_{cpo} \nabla_{\vartheta} (L^{\text{cpo},t}(\vartheta) + L^{\text{cpo},s}(\theta))$
- 14: $\phi \leftarrow \phi + \alpha_{cpo} \nabla_{\phi} L^{\text{value}}(\phi)$
- 15: **结束循环**
- 16: **对于** 轮次 $i = 0, 1, \dots$ **执行**
- 17: 使用MSE更新学生编码器：
- 18: $\theta_s \leftarrow \theta_s + \alpha_{mse} \nabla_{\theta_s} (L^s(\theta_s))$
- 19: **结束循环**

D. 高度渐进式分阶段奖励

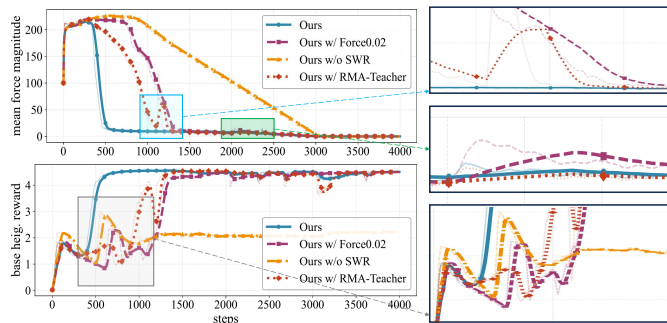
为了解决先前分阶段奖励设计中固定目标高度导致的训练不稳定性，我们采用高度递进式分阶段奖励，根据智能体级别的高度阈值逐步更新目标基准高度。最终的奖励函数及其权重列于表II。每一层采用不同的奖励集，以引导机器人逐步实现恢复：**(1) 上半身直立** r^u 。这一初始阶段旨在驱动机器人的上半身从俯卧姿态过渡到直立姿态，主要通过重力投影惩罚进行引导，目标基准高度设置为 h_1^{cmd} 。我们基于高度阈值定义智能体集合 $S_1 = \{i \mid h > h_1^{\text{cmd}}\}$ 和 $S_2 = \{i \mid h > h_2^{\text{cmd}}\}$ 。**(2)**

TABLE II
REWARD TERMS AND WEIGHTS

Reward	Definition	r^u	r^s	r^w
Lin. vel.	$\exp(-8.3\ v_{xy} - v_{xy}^{\text{cmd}}\ _2^2)$	-	-	7.0
Ang. vel.	$\exp(-8.3\ \omega_z - \omega_z^{\text{cmd}}\ _2^2)$	-	-	4.0
Orient.	$\ \mathbf{g}_{xy}\ _2$	-2.6	-2.8	-10.0
Torques	$\ \tau\ _2^2$	$-1.0e^{-6}$	$-1.0e^{-6}$	$-1.0e^{-6}$
DOF acc.	$\dot{q}^2$	$-2.5e^{-8}$	$-2.5e^{-6}$	$-2.5e^{-6}$
DOF vel.	$\dot{q}^2$	-	-	-0.01
Act. rate	$\ a_t - a_{t-1}\ _2^2$	-0.06	-0.02	-0.08
Act. smooth	$\ a_t - 2a_{t-1} + a_{t-2}\ _2^2$	-	-	-0.04
Dof pos.	$\ q - q_{\text{nominal}}\ _2^2$	-0.06	-0.1	-0.12
Dof ener.	$\ \dot{q}\tau\ _2^2$	-	-	$-1.0e^{-4}$
Base heig.	$\exp(-8.3\ h - h_i^{\text{cmd}}\ _2^2)$	4.0	6.0	5.0
Termin.	n^{te}	-200	-100	-500
Feet dist.	$\ p_y^{\text{lw}} - p_y^{\text{rw}} - 0.35 \ _2$	-	-	-2.0
Leg bias	$\ q^{\text{left}} - q^{\text{right}}\ _2$	-2.0	-0.02	-
No fly	n^{nf}	-2.0	-	-0.2
Wheel force	$\ f^{\text{wheel}} - M\ _2^2$	4.0	-	-

TABLE III
DOMAIN RANDOMIZATION

Parameter	Randomization Range	Unit
Base Mass	$\mathcal{U}(-0.1, 1.2)$	kg
Base CoM _x offset	$\mathcal{U}(-0.02, 0.02)$	m
Base CoM _y offset	$\mathcal{U}(-0.01, 0.01)$	m
Base CoM _z offset	$\mathcal{U}(-0.02, 0.02)$	m
Friction	$\mathcal{U}(0.1, 1.7)$	-
Restitution	$\mathcal{U}(0.3, 1.0)$	-
k_p and k_d	$\mathcal{U}(0.95, 1.05) \times \text{default}$	Nm/rad
Motor Strength	$\mathcal{U}(0.85, 1.05)$	-
Initial joint angle	$\mathcal{U}(0.5, 1.5) \times \text{default joint angle}$	-
Initial base euler	$\mathcal{U}(-0.3, 0.3)$	-

Fig. 3. **External force and height reward curves.** Smoother curves that reach their extremum (lowest/highest point) more rapidly indicate that the corresponding method produces a more stable policy and achieves faster training convergence.

A. Training Details

Model Initialization: Our method embeds the penalty-function approach within a global stochastic search framework, which typically requires extended training. Since RL performance often degrades in long-horizon tasks, direct training from random initialization may fail to converge. Rather than deepening the network, which can yield low-success-rate policies, we initialize training from a basic model pre-trained with r^w . Training stops once the robot achieves elementary walking motion, usually within about 200 iterations.

Terrain: To improve robustness and sim-to-real transfer, training is conducted across five terrain types: flat ground, slopes ($\leq 20^\circ$), pyramid steps (step height $\leq 0.12\text{m}$), discrete uneven terrain (height variation $\leq 0.12\text{m}$), and rough surfaces (roughness $\leq 0.03\text{m}$). This diversity encourages the policy to generalize across environments and helps bridge the sim-to-real gap induced by model discrepancies.

Domain Randomization: To bridge the sim-to-real gap, we applied domain randomization to key parameters during training (Table III), including PD gains, center-of-mass offset, torque offset, and initial offsets for joint position/velocity and base pose. The robot’s initial base pose was randomized across four states—front, down, or lying on either side—with an additional pose offset applied as perturbation.

B. Simulation Results

Under identical reward functions and hyperparameter settings, we conducted comparative experiments across the fol-

Standing-up r^s . This phase is activated when $|S_1| > \frac{2}{3}N$, i.e., when the number of agents whose base height exceeds h_i^{cmd} reaches two-thirds of the total population N . The objective is to encourage the robot to retract its lower body, bringing the leg joints back to angles close to the initial standing configuration to achieve a stable standing pose, and the target base height is updated to h_2^{cmd} . **(3) Walking r^w .** The final phase begins when $|S_2| > \frac{2}{3}N$, meaning the number of agents with $h > h_2^{\text{cmd}}$ reaches two-thirds of the total. This phase promotes normal locomotion after standing up, preventing the robot from remaining stationary, and the target base height is further updated to h_3^{cmd} . The symbol "-" denotes that the reward is inactive for that stage.

While the primary reward functions are adopted from [25], we specifically designed several additional reward terms (highlighted in red in Table II) to address key challenges:

- **Termination Penalty:** To prevent agents from lying idle on the ground, any agent contacting the ground for over 10 seconds is terminated, and a penalty is applied based on the number of terminated agents n^{te} .
- **Posture Penalties:** To penalize unnatural postures, we penalize aerial motion by counting agents with both wheels off the ground n^{nf} , penalize changes in leg spacing using the Y-coordinates of the left and right legs in the robot frame p_y^{lw} , p_y^{rw} , and penalize asymmetric leg poses by calculating the difference between left and right leg joint angles $q^{\text{left}} - q^{\text{right}}$.
- **Leg-Usage Reward:** o encourage leg usage during standing, we reward agents when wheel contact force f^{wheel} approaches total weight M .

III. EXPERIMENTS AND RESULTS

All training was conducted in IsaacGym [26] on an NVIDIA GeForce RTX 4070 Super GPU. We simulated 4,000 parallel environments (with 3,000 teacher and 1,000 student agents) for 8,000 iterations. The policy was trained at 200 Hz and updated at 50 Hz, while the hardware operated at 500 Hz to balance training speed and stability.

表 II 奖励项与权重

奖励	定义	r^u	r^s	r^w
线速度	$\exp(-8.3\ v_{x,y} - v_{x,y}^{\text{cmd}}\ _2^2)$	-	-	7.0
角速度	$\exp(-8.3\ \omega_z - \omega_z^{\text{cmd}}\ _2^2)$	-	-	4.0
方向	$\ \mathbf{g}_{xy}\ _2$	-2.6	-2.8	-10.0
力矩	$\ \tau\ _2^2$	$-1.0e^{-6}$	$-1.0e^{-6}$	$-1.0e^{-6}$
自由度加速度	$\dot{q}^2$	$-2.5e^{-8}$	$-2.5e^{-6}$	$-2.5e^{-6}$
自由度速度	$\dot{q}^2$	-	-	-0.01
动作频率	$\ a_t - a_{t-1}\ _2^2$	-0.06	-0.02	-0.08
动作平滑度	$\ a_t - 2a_{t-1} + a_{t-2}\ _2^2$	-	-	-0.04
自由度位置	$\ q - q_{\text{nominal}}\ _2^2$	-0.06	-0.1	-0.12
自由度能量	$\ \dot{q}\tau\ _2^2$	-	-	$-1.0e^{-4}$
基准高度	$\exp(-8.3\ h - h_i^{\text{cmd}}\ _2^2)$	4.0	6.0	5.0
终止	n^{te}	-200	-100	-500
足部距离	$\ p_y^{\text{lw}} - p_y^{\text{rw}} - 0.35 \ _2$	-	-	-2.0
腿部偏置	$\ q^{\text{left}} - q^{\text{right}}\ _2$	-2.0	-0.02	-
禁止飞行	n^{nf}	-2.0	-	-0.2
轮力	$\ f^{\text{wheel}} - M\ _2^2$	4.0	-	-

站立 r^s . 当 $|S_1| > \frac{2}{3}N$ 时，即基准高度超过 h_1^{cmd} 的智能体数量达到总群体 N 的三分之二时，该阶段被激活。其目标是鼓励机器人收回下半身，使腿部关节恢复到接近初始站立构型的角度，以实现稳定站立姿态，同时目标基准高度更新为 h_2^{cmd} . **(3) 行走 r^w .** 当 $|S_2| > \frac{2}{3}N$ 时，即具有 $h > h_2^{\text{cmd}}$ 的智能体数量达到总数的三分之二时，最终阶段开始。该阶段促进站立后的正常运动，防止机器人保持静止，目标基准高度进一步更新为 h_3^{cmd} . 符号“-”表示该奖励在该阶段不生效。

虽然主要奖励函数借鉴自 [25]，，我们专门设计了几个额外的奖励项（表II中红色高亮部分）以应对关键挑战：

- **终止惩罚：**为防止智能体在地面上闲置，任何与地面接触超过10秒的智能体将被终止，并根据被终止的智能体数量施加惩罚 n^{te} 。
- **姿态惩罚：**为了惩罚不自然的姿态，我们通过统计双轮离地的智能体数量来惩罚空中运动 n^{nf} ，通过机器人坐标系中左右腿的Y坐标来惩罚腿部间距变化 $p_{\text{lw}y}$, $p_{\text{rw}y}$ ，并通过计算左右腿关节角度之差来惩罚不对称腿部姿态 $q_{\text{左}} - q_{\text{右}}$ 。
- **腿部使用奖励：**为了鼓励站立时使用腿部，当轮地接触力 f^{wheel} 接近总重量 M 时，我们给予智能体奖励。

III. 实验与结果

所有训练均在IsaacGym中进行 [26]，使用NVIDIA GeForce RTX 4070 Super GPU。我们模拟了4,000个并行环境（其中3,000个教师智能体和1,000个学生智能体），共进行8,000次迭代。策略以200 Hz的频率训练，以50 Hz的频率更新，而硬件以500 Hz的频率运行，以平衡训练速度和稳定性。

表 III 域随机化

参数	随机化范围	Unit
基础质量	$\mathcal{U}(-0.1, 1.2)$	kg
基础质心 _x 偏移	$\mathcal{U}(-0.02, 0.02)$	m
基础质心 _y 偏移	$\mathcal{U}(-0.01, 0.01)$	m
基础质心 _z 偏移	$\mathcal{U}(-0.02, 0.02)$	m
摩擦	$\mathcal{U}(0.1, 1.7)$	-
恢复系数	$\mathcal{U}(0.3, 1.0)$	-
k_p 和 k_d	$\mathcal{U}(0.95, 1.05) \times \text{默认}$	牛米/弧度
电机强度	$\mathcal{U}(0.85, 1.05)$	-
初始关节角度	$\mathcal{U}(0.5, 1.5) \times \text{默认关节角度}$	-
初始底座欧拉角	$\mathcal{U}(-0.3, 0.3)$	-

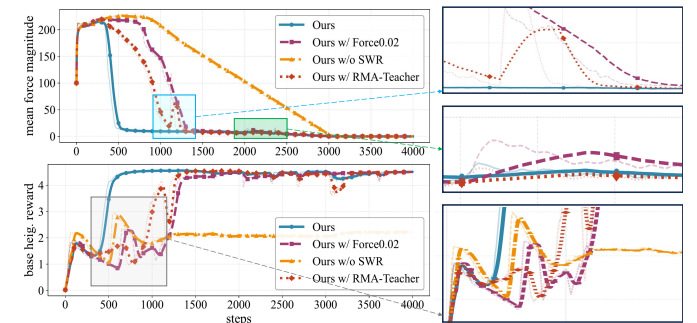


图3. 外力与高度奖励曲线。更平滑且更快达到极值（最低/最高点）的曲线表明相应方法能产生更稳定的策略，并实现更快的训练收敛。

A. 训练细节

模型初始化：我们的方法将惩罚函数方法嵌入到全局随机搜索框架中，这通常需要较长的训练时间。由于强化学习性能在长时域任务中往往会下降，从随机初始化直接训练可能无法收敛。我们不是通过加深网络来获得低成功率的策略，而是从使用 r^w 预训练的基础模型开始初始化训练。训练在机器人实现基本行走动作后停止，通常约在200次迭代内完成。

地形：为提高鲁棒性和仿真到现实迁移能力，训练在五种地形上进行：平坦地面、斜坡 ($\leq 20^\circ$)、金字塔台阶（台阶高度 $\leq 0.12\text{米}$ ）、离散不平地形（高度变化 $\leq 0.12\text{米}$ ）和粗糙表面（粗糙度 $\leq 0.03\text{米}$ ）。这种多样性鼓励策略在不同环境中泛化，并有助于弥合由模型差异引起的仿真到现实差距。

域随机化：为弥合仿真到现实差距，我们在训练期间对关键参数应用了域随机化（表III），包括PD增益、质心偏移、力矩偏移以及关节位置/速度和基座姿态的初始偏移。机器人的初始基座姿态在四种状态间随机化——前倾、下倾或侧躺于任一侧——并额外施加姿态偏移作为扰动。

B. 仿真结果

在相同的奖励函数和超参数设置下，我们在以下

TABLE IV
ABLATION STUDY RESULTS (MEAN $\pm$ STD).

Method	Ground				Slope				Step			
	$E_{\text{sacc}} \uparrow$	$E_{\text{feet}} \downarrow$	$E_{\text{smth}} \downarrow$	$E_{\text{engy}} \downarrow$	$E_{\text{sacc}} \uparrow$	$E_{\text{feet}} \downarrow$	$E_{\text{smth}} \downarrow$	$E_{\text{engy}} \downarrow$	$E_{\text{sacc}} \uparrow$	$E_{\text{feet}} \downarrow$	$E_{\text{smth}} \downarrow$	$E_{\text{engy}} \downarrow$
(a) Ablation on Height-progressive Stage-Wise Rewards												
Ours w/o SWR	0.0	/	/	/	0.0	/	/	/	0.0	/	/	/
Ours	99.8 (± 0.7)	0.52 (± 0.2)	5.80 (± 3.7)	4.36 (± 0.2)	99.5 (± 0.4)	0.68 (± 0.3)	7.05 (± 6.19)	5.01 (± 0.7)	99.8 (± 0.44)	0.53 (± 0.1)	6.83 (± 2.5)	4.12 (± 0.7)
(b) Ablation on Force-Guided												
Ours w/ Force Curr.	0.0	/	/	/	0.0	/	/	/	0.0	/	/	/
Ours w/o Force Guid.	97.4(± 2.7)	0.82(± 1.0)	12.4(± 3.5)	4.1 (± 0.3)	98.4(± 0.4)	0.42 (± 0.2)	12.75(± 2.84)	5.69(± 0.3)	100.0 (± 0.0)	0.3 (± 0.1)	12.5(± 2.8)	5.05(± 0.4)
Ours-Force0.02	87.8(± 3.6)	4.29(± 8.5)	12.28(± 4.8)	5.53(± 0.85)	99.5(± 0.8)	0.69(± 0.1)	10.77(± 4.9)	5.59(± 0.5)	99.0(± 0.4)	4.67(± 0.1)	8.48(± 6.9)	5.48(± 0.3)
Ours	99.8 (± 0.7)	0.52 (± 0.2)	5.80 (± 3.7)	4.36(± 0.2)	99.5 (± 0.4)	0.68(± 0.3)	7.05 (± 6.19)	5.01 (± 0.7)	99.8(± 0.44)	0.53(± 0.1)	6.83 (± 2.5)	4.12 (± 0.7)
(c) Ablation on Teacher-Student												
Ours w/o T-S	59.0(± 10.7)	1.5(± 0.6)	50.8(± 8.1)	5.52(± 0.5)	0.0	/	/	/	0.0	/	/	/
Ours w/ RMA Tea.	100.0 (± 0.0)	0.6(± 0.3)	4.80 (± 1.7)	5.36(± 0.12)	99.8 (± 0.7)	0.56 (± 0.4)	6.12 (± 4.23)	4.46 (± 0.7)	99.0(± 0.54)	0.71(± 0.1)	5.71 (± 1.5)	6.34(± 0.8)
Ours	99.8(± 0.7)	0.52 (± 0.2)	5.80(± 3.7)	4.36 (± 0.2)	99.5(± 0.4)	0.68(± 0.3)	7.05(± 6.19)	5.01(± 0.7)	99.8 (± 0.44)	0.53 (± 0.1)	6.83(± 2.5)	4.12 (± 0.7)

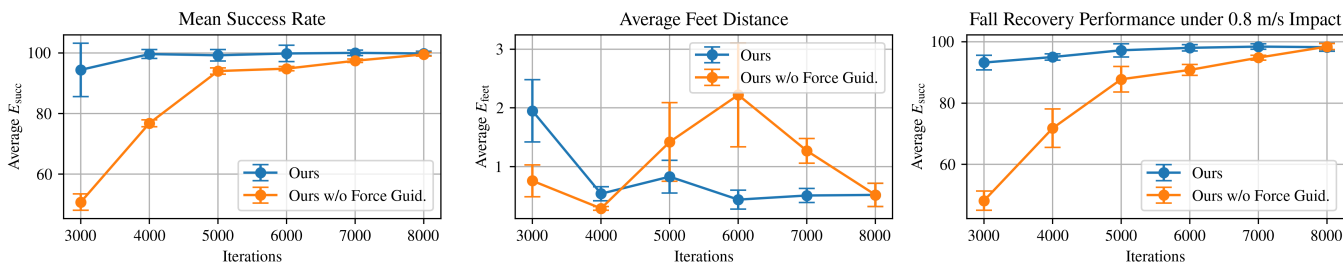


Fig. 4. **Success rate and foot travel distance across training iterations.** Compared with force-free baselines, our force-guided approach achieves a high recovery success rate by 3k iterations and attains improved stability by 4k iterations. These results demonstrate the rapid convergence and effectiveness of the proposed force-guided method.

lowing eight frameworks:

- **Ours:** Our full framework, which integrates force-guided with stage-wise rewards structure and a teacher-student training architecture.
- **Ours w/ Force Curr.:** Here, the force-guided is replaced with a force curriculum learning strategy [15], while keeping all other components unchanged.
- **Ours w/o Force Guid.:** In this variant, the explicit force-guided term is removed from the training process. The robot is still subjected to an external force, but without the force-guided method.
- **Ours w/ Force0.02.:** The baseline penalty multiplier is set to 0.001. To demonstrate that excessive Force-Guided leads to policy degradation, a comparative experiment is conducted with the multiplier increased to 0.02.
- **Ours w/o SWR:** This variant removes the height-progressive stage-wise rewards design, utilizing only a single-layer reward signal r^w .
- **Ours w/o T-S:** This variant abandons the teacher-student framework, reverting to a standard Actor-Critic structure.
- **Ours w/ RMA Tea.:** This variant corresponds to the teacher model in [27], which embodies the optimal outcome within the teacher-student learning framework.
- **HoST:** This refers to the method proposed in [15].

Our framework enables armless bipedal-wheeled robots to recover from falls while maintaining sustained locomotion, and further generalizes to a humanoid platform. Ablation studies using metrics from [15] evaluate each component

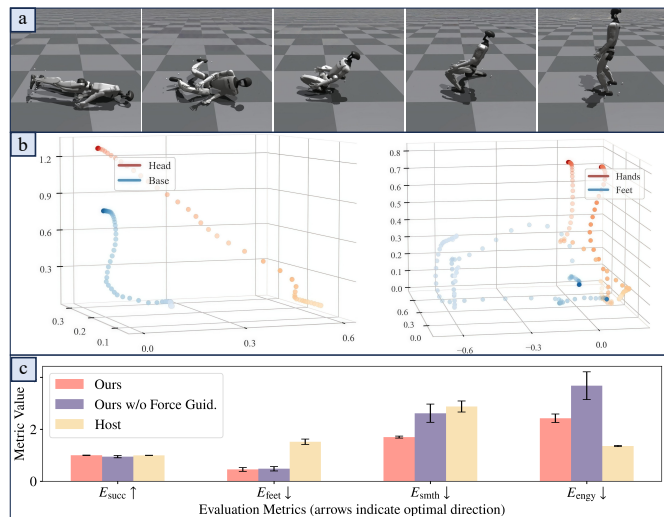


Fig. 5. **The proposed force-guided method is extended to a 23-DOF Unitree humanoid.** Our method learns a policy to stand up from arbitrary poses (a). The recovery motion is illustrated via 3D trajectories of key points (b). and a comparative evaluation (c) confirms the policy's smoothness and superiority over other methods.

across varied terrains, with results summarized as follows:

Height-progressive stage-wise rewards is crucial for guiding the recovery process. Without the initial reward layers and adaptive target height updates, the force guided policy may rise with assistance but collapses upon force removal due to poor posture—evident in the pronounced force-curve

表IV 消融研究结果 (均值 $\pm$ 标准差)。

方法	地面				斜坡				Step			
	$E_{\text{sacc}} \uparrow$	$E_{\text{feet}} \downarrow$	$E_{\text{smth}} \downarrow$	$E_{\text{engy}} \downarrow$	$E_{\text{sacc}} \uparrow$	$E_{\text{feet}} \downarrow$	$E_{\text{smth}} \downarrow$	$E_{\text{engy}} \downarrow$	$E_{\text{sacc}} \uparrow$	$E_{\text{feet}} \downarrow$	$E_{\text{smth}} \downarrow$	$E_{\text{engy}} \downarrow$
(a) 高度渐进式分阶段奖励的消融实验。												
我们的方法 (无SWR)	0.0	/	/	/	0.0	/	/	/	0.0	/	/	/
Ours	99.8 (± 0.7)	0.52 (± 0.2)	5.80 (± 3.7)	4.36 (± 0.2)	99.5 (± 0.4)	0.68 (± 0.3)	7.05 (± 6.19)	5.01 (± 0.7)	99.8 (± 0.44)	0.53 (± 0.1)	6.83 (± 2.5)	4.12 (± 0.7)
(b) 力引导消融实验												
我们的方法 (带电流)	0.0	/	/	/	0.0	/	/	/	0.0	/	/	/
我们的方法 (无力引导)	97.4(± 2.7)	0.82(± 1.0)	12.4(± 3.5)	4.1 (± 0.3)	98.4(± 0.4)	0.42 (± 0.2)	12.75(± 2.84)	5.69(± 0.3)	100.0 (± 0.0)	0.3 (± 0.1)	12.5(± 2.8)	5.05(± 0.4)
我们的方法-力0.02	87.8(± 3.6)	4.29(± 8.5)	12.28(± 4.8)	5.53(± 0.85)	99.5(± 0.8)	0.69(± 0.1)	10.77(± 4.9)	5.59(± 0.5)	99.0(± 0.4)	4.67(± 0.1)	8.48(± 6.9)	5.48(± 0.3)
Ours	99.8 (± 0.7)	0.52 (± 0.2)	5.80 (± 3.7)	4.36(± 0.2)	99.5 (± 0.4)	0.68(± 0.3)	7.05 (± 6.19)	5.01 (± 0.7)	99.8(± 0.44)	0.53(± 0.1)	6.83 (± 2.5)	4.12 (± 0.7)
(c) 师生消融实验												
无师生蒸馏的我们的方法	59.0(± 10.7)	1.5(± 0.6)	50.8(± 8.1)	5.52(± 0.5)	0.0	/	/	/	0.0	/	/	/
使用RMA Tea的我们的方法	100.0 (± 0.0)	0.6(± 0.3)	4.80 (± 1.7)	5.36(± 0.12)	99.8 (± 0.7)	0.56 (± 0.4)	6.12 (± 4.23)	4.46 (± 0.7)	99.0(± 0.54)	0.71(± 0.1)	5.71 (± 1.5)	6.34(± 0.8)
Ours	99.8(± 0.7)	0.52 (± 0.2)	5.80(± 3.7)	4.36 (± 0.2)	99.5(± 0.4)	0.68(± 0.3)	7.05(± 6.19)	5.01(± 0.7)	99.8 (± 0.44)	0.53 (± 0.1)	6.83(± 2.5)	4.12 (± 0.7)

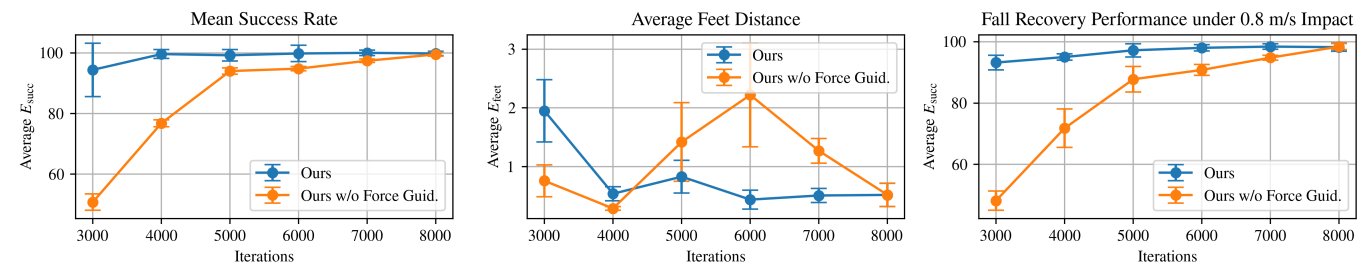


图4. 训练迭代中的成功率和足部移动距离。与无力基线相比，我们的力引导方法在3k次迭代时实现了高恢复成功率，并在4k次迭代时获得了更高的稳定性。这些结果证明了快速收敛和有效性

所提出的力引导方法的有效性。

八个框架中进行了对比实验：

- **我们的方法：** 我们的完整框架，将力引导与分阶段奖励结构和师生训练架构相结合。
- **使用力课程的我们的方法：** 此处，力引导被替换为力课程学习策略 [15]，，同时保持所有其他组件不变。
- **我们的方法（无力引导）：** 在此变体中，显式的力引导项从训练过程中被移除。机器人仍会受到外力作用，但不采用力引导方法。
- **我们的方法（力系数0.02）：** 基线惩罚系数设置为0.001。为证明过度的力引导会导致策略退化，我们进行了对比实验，将系数提高至0.02。
- **我们的方法（无SWR）：** 此变体移除了高度递进的分阶段奖励设计，仅使用单层奖励信号 r^w 。
- **无师生蒸馏的我们的方法：** 此变体放弃了师生框架，回归到标准的演员-评论家结构。
- **我们的方法（含RMA Tea）：** 此变体对应 [27] 中的教师模型，该模型体现了师生学习框架内的最优结果。

- **HoST：** 这是指 [15]中提出的方法。

我们的框架使无臂双足轮式机器人能够在保持持续运动的同时从跌倒中恢复，并进一步泛化到人形平台。使用来自 [15] 的指标的消融研究评估了每个组件

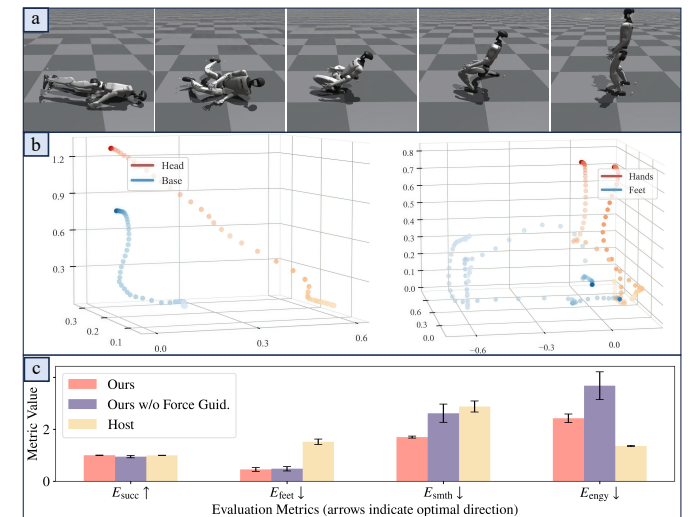


图5. 所提出的力引导方法已扩展至23自由度的宇树人形机器人。我们的方法学习了一种从任意姿态站立策略(a)。恢复运动通过关键点的3D轨迹展示(b)，对比评估(c)证实了该策略的平滑性及其相对于其他方法的优越性。

在多变地形上的表现，结果总结如下：

高度渐进式分阶段奖励对于引导恢复过程至关重要。 如果没有初始奖励层和自适应目标高度更新，力引导策略可能会在辅助下上升，但由于姿态不佳而在撤去外力后崩溃——这从明显的力曲线中可见一斑。

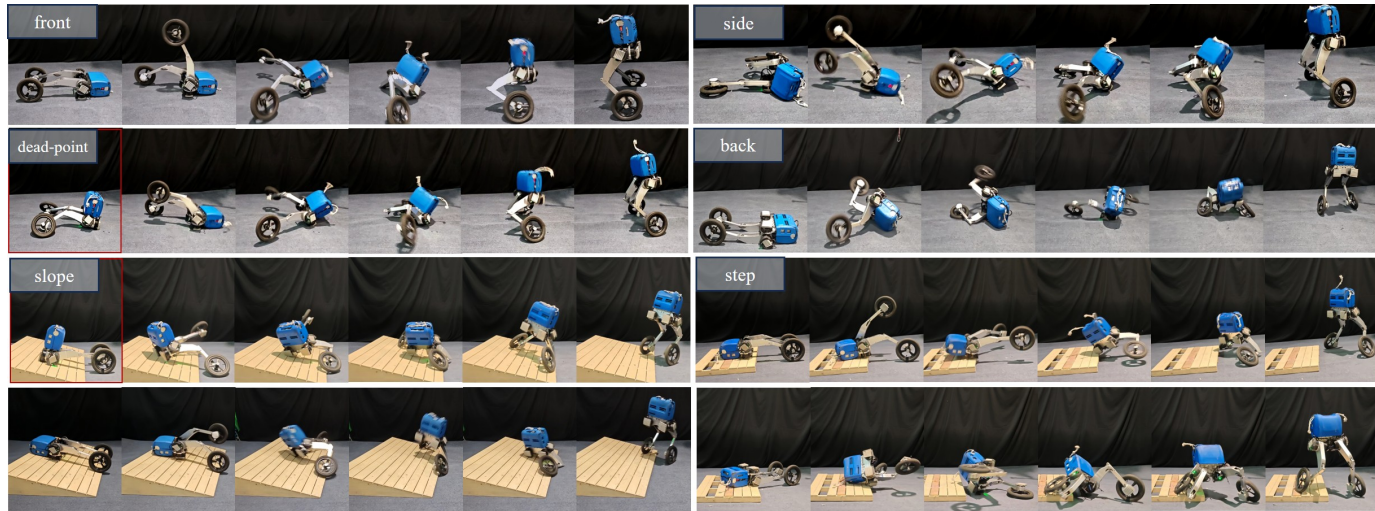


Fig. 6. **Standing-up tests under varied poses and terrains.** The dead-point pose is highlighted with a box, which represents the final posture achieved by the force-curriculum method. Our approach successfully overcomes this pose constraint and enables full standing recovery.

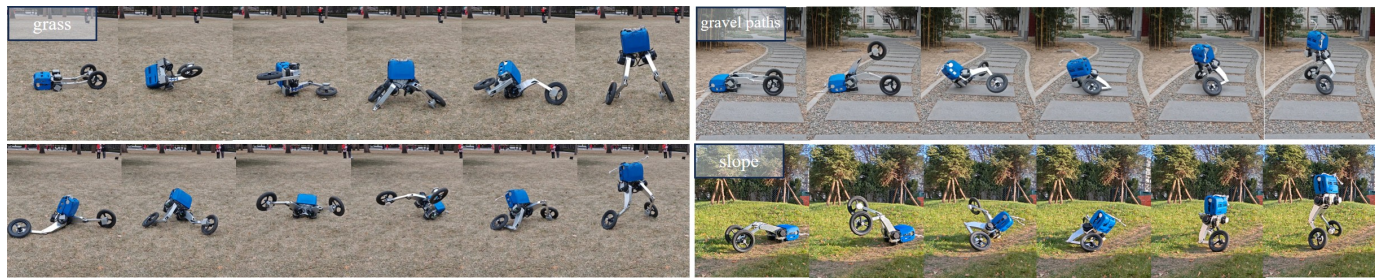


Fig. 7. **Standing-up tests outdoors under various initial poses.** These results demonstrate the robustness and environmental adaptability of our approach.

rebound during training (Fig. 3, Table. IV), reflecting slow and unstable convergence. In contrast, the stage-wise rewards policy attains stable locomotion at 0.8m/s and maintains a high recovery success rate while moving (Fig. 4).

Proper force-guided learning leads to more robust and efficient recovery strategies. The conventional force-curriculum approach fails to achieve standing, while our redesigned force function enables rising but yields poor metrics (Table IV). In contrast, our method formulates force as optimizable constraints, driving rapid convergence to low-intervention and robust regions. Assistance is removed after 3k iterations, with stable, rapid force decay (Fig. 3). The policy achieves force-free standing by 3k iterations, stabilizes after 4k with minimal foot adjustment, and maintains high success under commanded speeds (Fig. 4).

Privileged information delivered through the teacher-student architecture improves recovery success. Training with proprioception alone (**Ours w/o T-S**) yields limited results, while incorporating privileged observations via the teacher-student framework significantly enhances performance, achieving comparable results to **Ours w/ RMA Tea**. (Table IV). This demonstrates that the teacher-student architecture effectively distills privileged knowledge of force effects and recovery dynamics, thereby improving the overall robustness and adaptability of the recovery process.

The method also generalizes to a 23-DOF humanoid (Uni-

tree), producing stable, smooth recovery and showing better foot control and motion smoothness than HoST after only 8k iterations (Fig. 5), further validating the FTSR framework.

C. Real-World Experiments

To validate the effectiveness of our approach, we deployed the trained policy on our self-developed *JiaRan* robot and conducted real-world tests across various terrains and initial poses. All experimental results demonstrate the robustness and adaptability of our method.

We evaluated the policy’s standing recovery across a range of initial poses (front, back, side, and dead-point) and diverse terrains, including indoor flat ground, slope, and step (Fig. 6), as well as outdoor grass, slope, and gravel paths (Fig. 7). The policy demonstrated strong environmental adaptability and robustness throughout these tests. Furthermore, quantitative success rates across outdoor terrains are reported in Table V, collectively confirming the method’s consistency, stability, and reliable adaptation in complex real-world conditions.

To demonstrate that the policy retains locomotor capability after recovery, we introduced kicking disturbances to the robot on grass and recorded the complete motion sequence, shown in Fig. 8. It can be observed that after being pushed with substantial external force, the robot loses balance and falls, after which the policy autonomously drives it to recover a standing pose while preserving motion control ability thereafter.

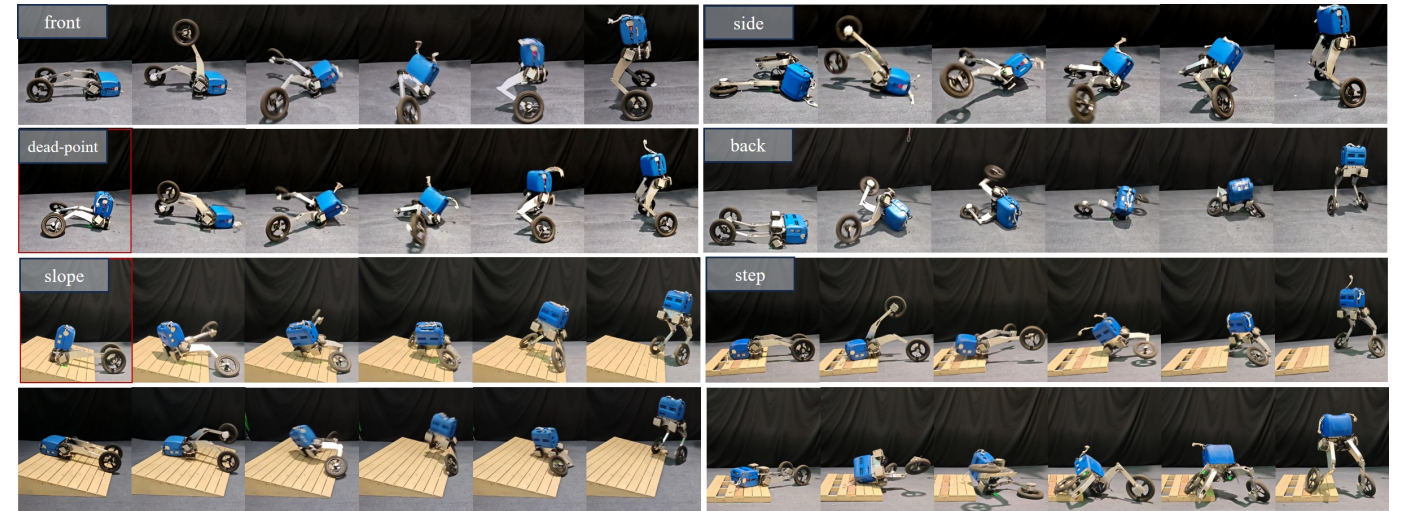


图6. 不同姿态和地形下的站立测试。死点姿态用方框标出，该姿态代表力课程方法最终达到的姿态。我们的方法成功克服了这一姿态约束，实现了完整的站立恢复。

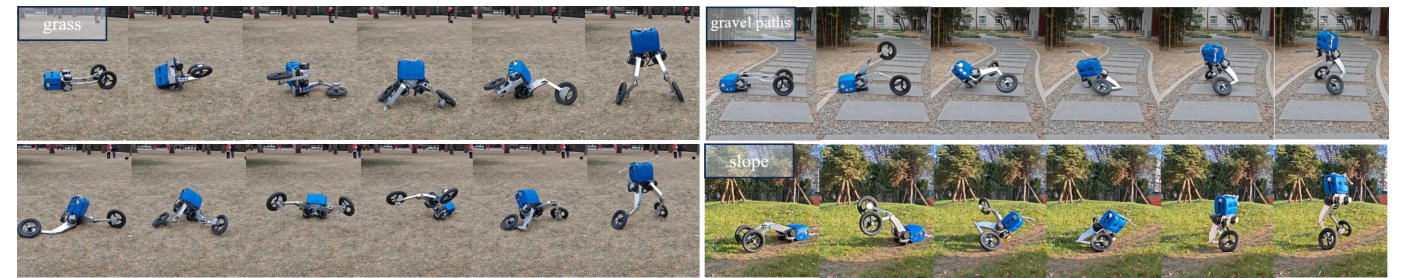


图7. 不同初始姿态下的户外站立测试。这些结果证明了我们方法的鲁棒性和环境适应性。

训练期间的反弹（图3、表IV）反映了缓慢且不稳定的收敛。相比之下，分阶段奖励策略在0.8m/s速度下实现了稳定运动，并在移动过程中保持了较高的恢复成功率（图4）。

适当的力引导学习能够带来更鲁棒、更高效的恢复策略。传统的力课程方法无法实现站立，而我们重新设计的力函数虽然能够实现起身，但指标表现不佳（表IV）。相比之下，我们的方法将力构建为可优化约束，驱动策略快速收敛到低干预且鲁棒的区域。辅助力在3k次迭代后移除，力衰减稳定且迅速（图3）。该策略在3k次迭代时实现无辅助站立，4k次迭代后趋于稳定且足部调整极小，并在指令速度下保持高成功率（图4）。

通过师生架构传递的特权信息提高了恢复成功率。仅使用本体感觉进行训练（无师生蒸馏的我们的方法）效果有限，而通过师生框架引入特权观测则显著提升了性能，取得了与**Ours w/ RMA Tea**相当的结果（表IV）。这表明师生架构有效地蒸馏了力效应和恢复动力学的特权知识，从而提高了恢复过程的整体鲁棒性和适应性。

树状结构），在仅8k次迭代后即可产生稳定、平滑的恢复动作，并在足部控制和动作平滑度上优于HoST（图5），进一步验证了FTSR框架的有效性。

C. 真实世界实验

为了验证我们方法的有效性，我们将训练好的策略部署到自主研发的嘉然机器人上，并在多种地形和初始姿态下进行了真实世界测试。所有实验结果都证明了我们方法的鲁棒性和适应性。

我们评估了策略在多种初始姿态（前倾、后倾、侧倾和死点）以及多样化地形下的站立恢复能力，包括室内平地、斜坡和台阶（图6），以及室外草地、斜坡和碎石路（图7）。该策略在这些测试中表现出强大的环境适应性和鲁棒性。此外，户外地形的定量成功率报告在表V中，共同证实了该方法在复杂真实世界条件下的稳定性、一致性和可靠适应性。

为了证明策略在恢复后仍保持运动能力，我们在草地上对机器人施加踢击扰动，并记录了完整的运动序列，如图8所示。可以观察到，在受到较大外力推动后，机器人失去平衡并摔倒，随后策略自主驱动其恢复站立姿态，并在此后保持运动控制能力。

该方 该方法同样适用于23自由度人形机器人（Uni-

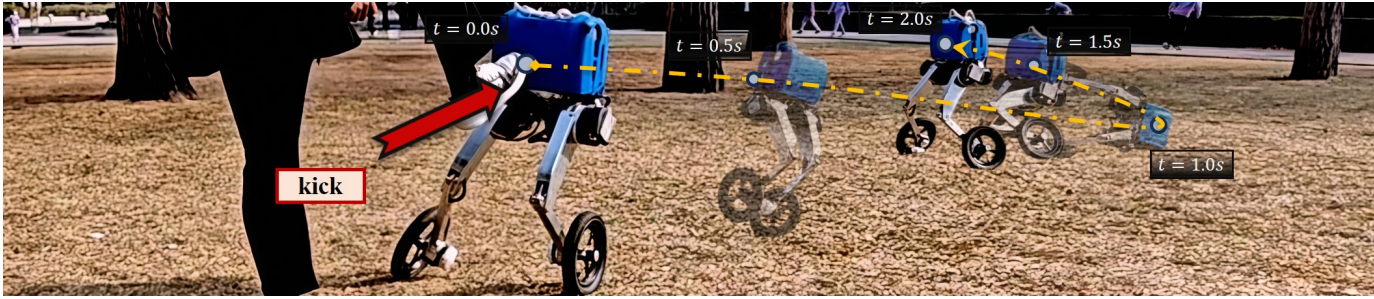


Fig. 8. **Fall recovery and subsequent locomotion test.** The direction of the external kick is marked with a red arrow, while the trajectory of the robot's base during the experiment is indicated by a yellow dashed arrow. The results demonstrate that our method can autonomously recover from a fall and continue to execute motion commands.

TABLE V
SUCCESS RATES ACROSS DIFFERENT OUTDOOR TERRAINS

Metric	Grass	Slope	Stairs	Overall
Success count / Total trials	5/5	5/5	5/5	15/15

IV. CONCLUSION

This letter proposes the Force-guided Teacher-student framework with Stage-wise Rewards (FTSR). The framework enables bipedal-wheeled robots to recover from falls across multiple randomized and diverse terrains. By integrating a force-guided method, it addresses the trajectory-free fall recovery problem, ensuring stability and higher training efficiency. The teacher-student architecture combined with height-progressive stage-wise reward shaping utilizes privileged information through phased training, producing robust, high-success-rate policies. Comprehensive experiments validate the method's robustness, adaptability, fast convergence, and stability on both simulated and physical bipedal-wheeled robots, also showing direct transfer potential to humanoids. Future work will test its utility on broader platforms.

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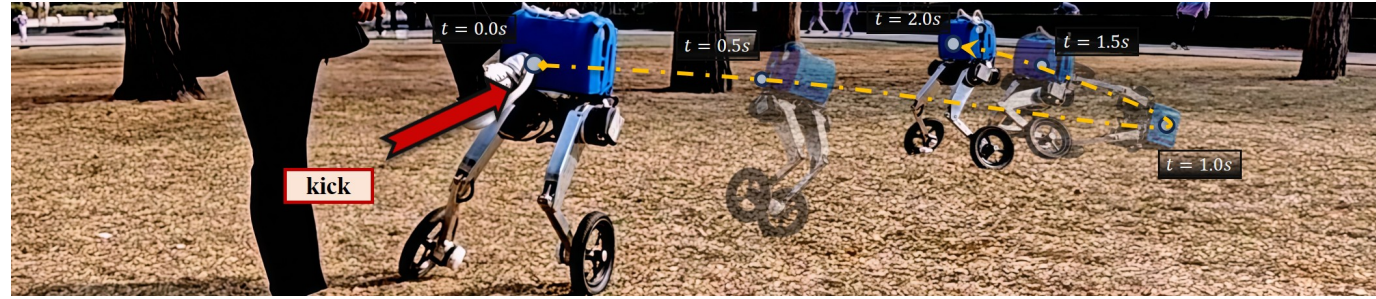


图8. 跌倒恢复及后续运动测试。外部踢击的方向用红色箭头标记，而实验过程中机器人基座的轨迹用黄色虚线箭头表示。结果表明，我们的方法能够自主恢复跌倒并继续执行运动指令。

表V 不同户外地形的成功率

指标	草地	斜坡	楼梯	总体
成功次数 / 总试验次数	5/5	5/5	5/5	15/15

IV. 结论

本文提出了力引导教师-学生分阶段奖励框架（FTSR）。该框架使双足轮式机器人能够在多种随机化和多样化地形上从摔倒中恢复。通过整合力引导方法，它解决了无轨迹摔倒恢复问题，确保了稳定性和更高的训练效率。教师-学生架构结合高度递进的分阶段奖励塑形，通过分阶段训练利用特权信息，生成鲁棒且高功率的策略。综合实验验证了该方法在仿真和实体双足轮式机器人上的鲁棒性、适应性、快速收敛性和稳定性，并展示了直接迁移到人形机器人的潜力。未来工作将测试其在更广泛平台上的实用性。

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